

An Uncertainty-Aware Cascaded Framework for Real-Time QR-Embedded URL Phishing Detection Under Deployment Constraints

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Abstract—The increasing use of QR codes in payments, authentication, and information access has led to QR-based phishing, or quishing, where malicious URLs are embedded in QR codes and executed upon scanning. This type of attack bypasses conventional browser based defenses. Real-time detection is necessary, but lightweight classifiers that could be used at the time of scanning have structural recall limitations against advanced phishing infrastructure. So we present an uncertainty aware cascaded detection framework that directs extracted URLs from QR codes through three levels of deep analysis. Tier-1 uses a calibrated XG-Boost classifier on a 17-dimensional lexical feature vector; Isotonic Regression calibration probabilities drive an uncertainty-based escalation threshold. Tier-2 does O(1) in memory root domain lookup against a self-curating proprietary blacklist where every Tier-3 confirmed malicious domain is automatically appended for sub millisecond re-detection of repeat threats without model retraining. Tier-3 uses weighted forensic scoring function over ten structural and behavioral signals with soft fail guarantee which ensures no user gets blocked if inspection fails. Tested on a set 80/20 split of 8,411,061 URLs, Tier-1 gets 90.84% accuracy, 73.19% recall with an AUC-ROC of 0.9304 at a high-confidence precision of 97.90% and escalation rate at 36.47%. Live testing on 200 known phishing URLs shows a reachability rate of 9.5%, with recall at 73.68% for reachable ones. More importantly, we show that the soft-fail guarantee letting unreachable URLs go as SAFE UNVERIFIED has actual recall cost under evasion-capable phishing infrastructure; it gives a predicted cascade recall of 68.57%. We define this soft fail recall tradeoff as an unquantified deployment challenge and suggest SUSPICIOUS UNVERIFIED as a confidence-weighted mitigation for future research endeavors.

Index Terms—URL phishing detection, uncertainty quantification, probability calibration, cascaded detection framework, QR code security, lexical feature analysis, threat intelligence, real-time security systems

I. INTRODUCTION

QR codes are now a key part of modern mobile ecosystems. They let people interact with each other without touching anything, and they can be used for digital payments, authentication, ticketing, and service access. Because they can't see the encoded content before scanning, they are a popular target for phishing attacks. Malicious QR codes send users to credential harvesting portals, fake payment gateways, or malware distribution pages with just one scan. The user has no chance to verify the link before clicking. The number of reported incidents of this type of attack, known as quishing, has grown a lot, and it is a structurally different threat from traditional hyperlink-based phishing [2], [3]. The current defenses are not strong enough to protect against this threat model. Commercial QR scanners mostly use static blacklists and signature matching, which are both reactive and don't protect against phishing domains that are new. Machine learning methods that use lexical and structural URL features have shown better generalization. However, lightweight models that can be used on mobile devices in real time have trouble recalling structures, usually only getting 70–75% of the way there [5]. Deep learning models get better results, but they take too long to work with interactive scan workflows. No currently deployed system addresses this tradeoff through principled uncertainty-aware routing; instead, they apply uniform analysis depth to all URLs, irrespective of classification confidence. It is important to note that no previous research has examined the recall cost associated with soft-fail guarantees when implemented against actual phishing infrastructure.

This paper presents an uncertainty-aware cascaded detection framework for identifying phishing URLs derived from QR codes. The main idea behind the architecture is that not all

URLs need the same level of analysis. A calibrated classifier can find safe URLs with high confidence in less than 10 milliseconds and only send samples that are not sure to more expensive inspection stages. Tier-1 uses an XGBoost classifier on a 17-dimensional lexical feature vector that has been calibrated through Isotonic Regression to give well-ordered probabilities. A single escalation threshold, T_{safe} , sends high-confidence safe predictions to immediate resolution and sends all uncertain and possibly malicious samples to Tier-1. Tier-1 never classifies a URL as malicious, which eliminates the risk of false negatives at the first stage. Tier-2 does a $O(1)$ root-domain lookup against a self-curating proprietary blacklist, which means that every domain is confirmed. Tier-3 automatically adds malicious information, which makes it possible to re-detect repeat threats in less than a millisecond without having to retrain the model. Tier-3 uses a weighted forensic scoring function on 21 structural and behavioral signals. A soft-fail guarantee makes sure that the system defaults to `SAFE_UNVERIFIED` instead of blocking a user when an inspection fails. This is because safety during deployment is more important than aggressive blocking. The framework doesn't need any outside threat intelligence APIs to work, so it can be used in networks with limited access. In addition to system performance, this work uncovers a previously unidentified deployment challenge: real-time testing against 200 confirmed phishing URLs shows that 90.5% of active phishing sites cannot be accessed by automated HTTP inspection because their domains have short lifespans, they have mechanisms for detecting bots, and they limit the number of requests they receive. Under the soft-fail guarantee, these unreachable URLs are released as `SAFE_UNVERIFIED`, revealing a fundamental tension between production safety and recall that has not been quantified in previous cascaded detection literature.

This work's main contributions are as follows:

- **Asymmetric safe-only Tier-1:** Calibrated XGBoost over 17 lexical features achieving 90.84% accuracy, 0.9304 AUC-ROC, and 97.90% high-confidence precision on 1,682,213 test URLs.
- **Calibration-driven uncertainty routing:** Isotonic Regression calibration partitions traffic into confident-safe (51.72%) and uncertain (36.47%) zones with a 6.94% fatal bypass rate.
- **Self-curating threat intelligence loop:** Tier-2 blacklist $\mathcal{B}$ grows via $\mathcal{B}_{t+1} = \mathcal{B}_t \cup \{D \mid \text{Tier-3}(U) = \text{MALICIOUS}\}$, enabling sub-millisecond re-detection without retraining.
- **Weighted forensic scoring:** Tier-3 aggregates 21 signals via $\text{Score}(U) = \sum_i w_i s_i$, achieving 73.68% recall on reachable live URLs at 1,663 ms median latency.
- **Soft-fail recall tradeoff:** Live evaluation quantifies 90.5% `SAFE_UNVERIFIED` rate against active phishing infrastructure, yielding 68.57% projected cascade recall a previously un-characterized deployment constraint.
- **Large-scale evaluation:** 8,411,061 URLs across predefined 80/20 stratified split, cross-validated against 200 live Open-Phish URLs.

The remainder of this paper is structured as follows. Section II reviews related work on phishing URL detection and QR code security. Section III describes the system architecture and formal specification. Section IV presents experimental results and tradeoff analysis. Section VI concludes with directions for future work.

II. RELATED WORK

A. URL-Based Phishing Detection

Traditional classifiers including Logistic Regression, Naive Bayes, SVM, and Random Forest have demonstrated strong baseline performance on lexical and structural URL features [6], [7]. XGBoost has shown consistently high accuracy on large-scale structured URL data [8], and deep learning models have reported accuracy exceeding 98% [9]. However, all of these approaches apply uniform analysis depth to every URL regardless of prediction confidence, with no mechanism to selectively escalate uncertain samples.

B. Multi-Stage Detection Pipelines

Several multi-stage architectures have been proposed. Qasem et al. [10] introduced a two-layer pipeline that classifies URLs then categorizes confirmed threats, but applies both stages uniformly to all inputs. A four-stage system integrating whitelist filtering, rule-based heuristics, ML classification, and Google Safe Browsing API lookup [11] is structurally closest to ours; however, it relies on an external commercial API as its final authority a hard dependency our framework eliminates entirely. A blacklist-and-XGBoost hybrid [12] places blacklist lookup before ML screening; our design deliberately inverts this ordering, using uncertainty-aware ML to eliminate high-confidence safe URLs before blacklist or forensic lookup.

C. QR Code Phishing Detection

Trad and Chehab [4] introduced a QR code analysis method based on visual and structural features, achieving an AUC of approximately 0.91. Their approach works directly on the pixel structure without extracting URLs. In contrast, URL-extraction methods [5] perform lexical analysis on decoded URLs but do not incorporate multi-layer verification. Both methods fail to cover the entire spectrum of possible attacks. Additionally, commercial scanners typically depend on static blacklists, which do not provide protection against zero-day domains.

D. Uncertainty Quantification

Probability calibration using Isotonic Regression and Platt Scaling has been studied for general binary classification [13]. Its use as a cascaded routing mechanism, where calibrated probabilities divide traffic into confident-safe and uncertain classes, has not been examined in phishing detection literature.

E. Positioning

Table I summarizes the key characteristics of previous systems in comparison to our framework. None of the existing systems simultaneously meet all five criteria, and no prior research has quantified the recall cost associated with soft-fail guarantees in real-world phishing infrastructures.

TABLE I
COMPARISON WITH RELATED SYSTEMS

System	Multi-Tier	Uncert. Routing	Self-Curating	No Ext. API	QR Native
Qasem et al. [10]	✓	×	×	✓	×
4-Stage [11]	✓	×	×	×	×
BL+XGB [12]	✓	×	×	✓	×
Trad et al. [4]	×	×	×	✓	✓
Calibration [13]	×	✓	×	✓	×
Proposed	✓	✓	✓	✓	✓

III. PROPOSED METHODOLOGY

The framework is implemented as a Flask backend with Redis for asynchronous task queues and Web-Socket communication to a mobile client. Tier-2 blacklist is Firebase backed. It is loaded into server RAM on startup. Fig. 1 shows the overall system and routing flow.

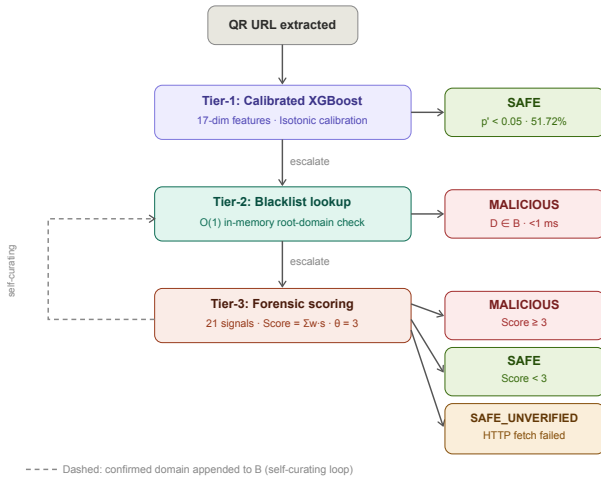


Fig. 1. Uncertainty-aware cascaded QR phishing detection framework.

A. System Overview

Given URL U extracted from a scanned QR code:

$$\text{Output}(U) = \begin{cases} \text{SAFE} & P'(U) < T_{\text{safe}} \\ \text{MALICIOUS} & \text{root}(U) \in \mathcal{B} \\ \text{MALICIOUS} & \text{Score}(U) \geq \theta \\ \text{SAFE} & \text{Score}(U) < \theta \\ \text{SAFE_UNVERIFIED} & \text{Tier-3 fetch failure} \end{cases} \quad (1)$$

where $P'(U)$ is the calibrated Tier-1 probability after deterministic override, $\mathcal{B}$ is the Tier-2 blacklist, and $\text{Score}(U)$ is the Tier-3 weighted forensic score. Tier-1 never issues MALICIOUS all non-safe traffic is escalated, preventing first-stage false negatives structurally.

B. Tier-1: Calibrated Lexical Classifier

Each URL is mapped to a 17-dimensional lexical feature vector $\mathbf{x} \in \mathbb{R}^{17}$ without making any network request. All features are listed in Table II. We use Shannon entropy $H(U) = -\sum_c p(c) \log_2 p(c)$ to measure URL randomness, and brand domain mismatch (Feature 16), which occurs when a protected brand appears in the path, but it is not the registered domain stem.

TABLE II
17-DIMENSIONAL LEXICAL FEATURE VECTOR

Idx	Feature	Type
1	URL length	Numeric
2	Dot count	Numeric
3	Hyphen count	Numeric
4	At-sign count	Numeric
5	Question mark count	Numeric
6	Ampersand count	Numeric
7	IP address presence	Binary
8	Digit count	Numeric
9	URL path depth	Numeric
10	Subdomain count	Numeric
11	Suspicious TLD	Binary
12	Shannon entropy	Numeric
13	Suspicious keyword count	Numeric
14	Max consecutive consonants	Numeric
15	Brand name in path	Binary
16	Brand domain mismatch	Binary
17	Longest URL token length	Numeric

If either Feature 11 or Feature 16 is active, a deterministic override guarantees $P'(U) = \max(P(U), 0.50)$, protecting structurally suspicious URLs from being cleared by Tier-1. The raw XGBoost scores are calibrated using Isotonic Regression [13] over 5-fold cross-validation, which makes $T_{\text{safe}} = 0.05$ a probabilistically meaningful safety boundary. URLs with $P'(U) < 0.05$ are resolved at sub-10ms; others are promoted to Tier-2.

C. Tier-2: Self-Curating Threat Intelligence

Escalated URLs undergo $O(1)$ root-domain lookup against in-memory blacklist $\mathcal{B}$. Tier-3 confirmations feed back into $\mathcal{B}$ autonomously:

$$\mathcal{B}_{t+1} = \mathcal{B}_t \cup \{ D \mid \text{Tier-3}(U) = \text{MALICIOUS}, D = \text{root}(U) \} \quad (2)$$

This self-curating loop lets you find repeat-offender domains again in less than a millisecond without having to retrain or rely on an outside API. URLs that don't match go up to Tier-3.

D. Tier-3: Weighted Forensic Scoring

Tier-3 does a live HTTP fetch and gets domain, SSL, redirect, and HTML signals at the same time using `ThreadPoolExecutor`. The score for risk is:

$$\text{Score}(U) = \sum_{i=1}^N w_i \cdot s_i, \quad \text{decision: MALICIOUS iff } \text{Score}(U) \geq \theta = 3 \quad (3)$$

where $s_i \in \{0, 1\}$ and w_i is the signal penalty weight. Table III shows the main types of signals. If the HTTP fetch fails, Tier-3 immediately returns `SAFE_UNVERIFIED`, which means that no user will be blocked if the inspection fails.

TABLE III
TIER-3 FORENSIC SIGNAL CATEGORIES

Category	Signal	$w_{\max}$
Brand Impersonation	Brand mismatch on login form	5
Hidden Traps	Hidden sensitive input fields	4
SSL Anomaly	No SSL on login page	4
Brand (no login)	Brand on unauthorized domain	4
Self-Signed Cert	Self-signed SSL certificate	3
Domain Age	Young domain with login/evasion	3
Form Action	Login submits to external domain	3
JS Obfuscation	Obfuscation on mismatched login	3
Redirect Evasion	Multi-domain redirect chain	2
Infrastructure	Bare IP host, suspicious TLD	2

IV. EXPERIMENTAL RESULTS

A. Dataset

The dataset contains 8,411,061 URLs taken from a community-created security corpus [16]. It is split into two groups: 6,728,848 URLs for training and 1,682,213 URLs for evaluation. The full dataset has 1,807,092 phishing (21.5%) and 6,603,969 legitimate (78.5%) examples. In both splits, the class ratio stays the same at 1:3.65. There are no network requests in the training set, so there is no live-data leakage. All features are taken directly from the URL string.

B. Tier-1 Standalone Performance

Table IV shows the Tier-1 metrics that were tested on a set of 1,682,213 URLs that were not used for training, with a hard classification threshold of 0.5. To fix the class imbalance, set `scale_pos_weight` to 3.66, which is the ratio of negative to positive examples in the training set.

C. ROC Analysis and Operating Point Selection

Fig. 2 shows the ROC curve for the calibrated Tier-1 classifier evaluated on the 1,682,213-URL test set. The model achieves AUC-ROC of 0.9304, confirming strong discriminative capability across all classification thresholds.

At the operating point of $T_{\text{safe}} = 0.05$, the TPR is 0.9306 and the FPR is 0.3602. This threshold has a high FPR on

TABLE IV
TIER-1 STANDALONE CLASSIFICATION PERFORMANCE

Metric	Value
Accuracy	90.84%
Precision	82.21%
Recall	73.19%
F1 Score	77.44%
AUC-ROC	0.9304
High-confidence precision	97.90%
Legitimate FP rate	0.32%

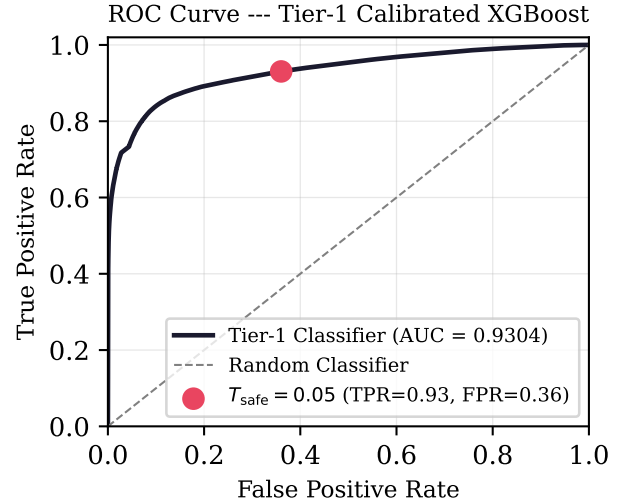


Fig. 2. ROC curve for Tier-1 calibrated XGBoost classifier (AUC = 0.9304). The red marker indicates the operating point at $T_{\text{safe}} = 0.05$ (TPR = 0.9306, FPR = 0.3602), selected to maximize phishing coverage at Tier-1 while deferring all uncertain samples to upstream tiers.

purpose and is acceptable from an architectural point of view. False positives at Tier-1 are not misclassifications; instead, they are escalated to Tier-2 and Tier-3 for resolution instead of being incorrectly blocked. The asymmetric design changes Tier-1 false positives from user-facing errors to computational costs. At this operating point, the 6.94% fatal bypass rate is equal to the TPR gap from 1.0, which means that phishing URLs have lexical features that are impossible to tell apart from legitimate traffic at the $p = 0.05$ boundary. Moving T_{safe} to the right lowers bypasses but raises escalation load; the chosen value is the best balance for mobile scan latency limits.

D. Uncertainty-Based Routing Analysis

Table V reports the traffic distribution across routing zones under thresholds $T_{\text{safe}} = 0.05$ and $T_{\text{mal}} = 0.85$ on the 1,682,213-URL test set.

TABLE V
TIER-1 ROUTING ZONE BREAKDOWN

Zone	URLs	% Traffic	Phishing %
SAFE ($p < 0.05$)	870,111	51.72%	6.94% bypass
ESCALATION ($0.05 \leq p \leq 0.85$)	613,426	36.47%	39.24% of phishing
HIGH-CONF ($p > 0.85$)	198,676	11.81%	97.90% precision

The SAFE zone clears 63.98% of real URLs right away, with a latency of less than 10 ms. The 6.94% fatal bypass rate means that phishing URLs with $p < T_{\text{safe}}$ are assigned by Tier-1 samples that don't get caught. The high-confidence zone has a 97.90% accuracy rate and only a 0.32% false positive rate on real traffic.

E. Tier-3 Live Evaluation

Tier-3 was evaluated on 200 verified active phishing URLs from the OpenPhish live feed [17]. Results are reported in Table VI.

TABLE VI
TIER-3 LIVE EVALUATION ON OPENPHISH URLS ($n = 200$)

Metric	Value
Reachable URLs	19 (9.5%)
SAFE_UNVERIFIED	181 (90.5%)
MALICIOUS correctly flagged	14
Missed (SAFE verdict)	5
Recall (reachable URLs)	73.68%
Latency mean	1,835 ms
Latency median	1,663 ms
Latency P95	3,732 ms
Avg. signals per phishing URL	1.50

Brand-domain mismatch (19.0%), WHOIS failure (10.5%), young domain (9.5%), and lack of SSL on login pages (7.0%) were the most common signals across confirmed phishing URLs. The average of 1.50 signals per confirmed phishing URL shows that most detections are based on high-weight single signals rather than convergent multi-signal evidence. This is because the scoring model relies on strong individual indicators like brand mismatch ($w = 5$) and hidden form fields ($w = 4$).

F. Soft-Fail Recall Tradeoff

The 90.5% SAFE_UNVERIFIED rate shows a basic problem with deployment. Short domain lifespans, bot-detection mechanisms, and aggressive rate limiting are all parts of modern phishing infrastructure. This makes it impossible for automated HTTP inspection to reach most active phishing URLs. The soft-fail guarantee releases unreachable URLs instead of blocking users. This means that the projected cascade recall is:

$$\begin{aligned} \text{Recall}_{\text{cascade}} &= \frac{|\text{esc_rescued}| + |\text{high_rescued}|}{|\text{phishing}_{\text{test}}|} \\ &= \frac{\lfloor 141,821 \times 0.7368 \rfloor + \lfloor 194,511 \times 0.7368 \rfloor}{361,419} = 68.57\% \end{aligned} \quad (4)$$

This falls 4.62 percentage points below the Tier-1 standalone baseline of 73.19%, as shown in Table VII.

This result reveals an unrecognized tension in cascaded detection design: the soft-fail guarantee that ensures system safety for production deployment simultaneously serves as

TABLE VII
SYSTEM PERFORMANCE COMPARISON

Metric	Tier-1 Only	Full Cascade
Accuracy	90.84%	—
Recall	73.19%	68.57% [†]
Precision	82.21%	97.90% [‡]
F1 Score	77.44%	—
AUC-ROC	0.9304	—
Escalation Rate	—	36.47%
Tier-3 Reachability	—	9.5%
SAFE_UNVERIFIED Rate	—	90.5%

[†] Projected using measured Tier-3 oracle rate of 73.68%

[‡] High-confidence zone only

a deterrent against cascade recall in the context of evasion-capable phishing infrastructure. Previous research on cascaded phishing detection assesses Tier-3 reachability implicitly using static datasets in which all URLs are resolvable, a condition that is not applicable in live deployment. We suggest using SUSPICIOUS_UNVERIFIED as a confidence-weighted intermediate state for future work. URLs that are raised with a lot of Tier-1 suspicion and then soft-fail would go into a quarantine state instead of being silently released. This keeps the no-block guarantee while showing the user that there is uncertainty about the detection.

V. DISCUSSION

A. Soft-Fail Recall Tradeoff

The 90.5% SAFE_UNVERIFIED rate shows a basic conflict in the design of deployment-aware detection. Static dataset evaluations take for granted that 100% of Tier-3 is reachable, which is not the case in a live deployment. Modern phishing infrastructure is hard to check automatically because it has short domain lifespans, bot-detection headers, and rate limiting. This means that the soft-fail guarantee is both a production necessity and a recall liability. The anticipated cascade recall of 68.57%, which is below the 73.19% Tier-1 standalone baseline, does not indicate a system failure; rather, it represents a genuine assessment of this tradeoff that previous literature on cascaded detection has not documented.

B. Proposed Mitigation and Future Directions

The SUSPICIOUS_UNVERIFIED state directly addresses this tradeoff: URLs with high Tier-1 suspicion ($p > T_{\text{high}}$) that encounter soft-fail are quarantined instead of being released without notice, which makes the user aware of the uncertainty without breaking the no-block guarantee. The self-curating blacklist is a way to make things better as $|B_t|$ grows over time. As the deployment time goes on, more and more repeat-offender domains are resolved at Tier-2 before they reach Tier-3, which lowers the risk of soft-fail over time. Future efforts will focus on browser-emulation fetch strategies to enhance Tier-3 accessibility, empirical validation of

SUSPICIOUS_UNVERIFIED, and longitudinal assessment of the blacklist growth rate.

VI. CONCLUSION AND FUTURE WORK

This paper presented an uncertainty-aware cascaded framework for real-time detection of phishing URLs extracted from QR codes. The system routes traffic through three tiers of progressively deeper analysis: a calibrated XGBoost lexical classifier with Isotonic Regression uncertainty routing, an $O(1)$ self-curating blacklist, and a weighted forensic scoring engine aggregating 21 behavioral and structural signals. Evaluated on 8,411,061 URLs across a pre-defined stratified split, Tier-1 achieves 90.84% accuracy, 73.19% recall, and 0.9304 AUC-ROC, with 97.90% precision in its high-confidence zone and a 36.47% escalation rate.

The main point of this study is not about improving recall but about a constraint on deployment: * live testing with 200 confirmed phishing URLs shows a 90.5% SUSPICIOUS_UNVERIFIED rate. This is because modern phishing sites have lifespans use bot-detection methods and limit the number of requests. The fail guarantee that stops users from being blocked if the inspection fails results in a predicted recall of 68.57%, which is 4.62% lower than the Tier-1 baseline. This tradeoff between fail and recall has not been measured in previous studies on cascaded detection. These studies assume that all data is reachable which is not the case, in real-world scenarios. We identified this gap. Suggest using SUSPICIOUS_UNVERIFIED as a quarantine state that weights confidence to regain recall without breaching the no-block deployment guarantee. This approach can help recover recall without compromising the safety of users.

Future work will focus on three areas. First we need to test the mechanism in real-life situations to see how well it works. We want to know if it can really detect activity. Second we need to improve how we reach Tier-3. We will try using browser- methods to fetch data and adjust our timeout rules to reduce failures caused by bot-detection systems. Third we plan to study how our self-updating blacklist grows over time as we use it. We want to see if the Tier-2 feedback loop really helps us learn and improve our system. We will track the growth of $|\mathcal{B}_t|$.

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